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Q1) Write a program to read a file and count how many vowels are present in its content.

Code:

```
with open("text.txt",'r') as f:
    content = f.read()
    print(content)
vowels = "aeiouAEIOU"
vowels_count = 0
for ch in content:
    if ch in vowels:
        vowels_count += 1
print(f"Total vowels {vowels_count}")
```

Output:

```
PS D:\Python> python -u "d:\Python\que1.py"
AtharvDugani
Total vowels 5
PS D:\Python>
```

Q2) Read all lines from a file into a list. Print the total number of lines and display the 2nd and 4th line

Code:

```
with open("text.txt", "r") as file:
    lines = file.readlines()
    print(lines)
print("Number of lines:", len(lines))
print(f"2nd line: {lines[1]}")
print(f"3rd line: {lines[2]}")
```

Output:

```
PS D:\career\Python> python -u "d:\career\Python\que2.py"
['Atharv Dugani\n', 'Today we will learn python\n', 'in that we will see file handling']
Number of lines: 3
2nd line: Today we will learn python

3rd line: in that we will see file handling
```

Q3) Take a sentence from the user, split it into words, and store them in a list. Print the longest word.

Code:

```python
s = input("Enter a sentence: ")
with open("data.txt", "w") as file:
    file.write(s)
with open("data.txt", "r") as file:
    words = file.read().split()
longest_word = ""
for word in words:
    if len(word) > len(longest_word):
        longest_word = word
print("Longest word:", longest_word)
```

Output:

```
PS D:\career\Python> python -u "d:\career\Python\que3.py"
Enter a sentence: Hello I am Atharv
Longest word: Atharv
PS D:\career\Python>
```

Q4) Read a file, split its content into words, and store them in a list.

i. Print the total number of words.
ii. Print all unique words.

Code:

```python
with open("text.txt", "r") as file:
    content = file.read()
    words = content.split()

    print(f"Total number of words: {len(words)}")
unique_words = set(words)
print("Unique words:", unique_words)
```

Output:

```
PS D:\career\Python> python -u "d:\career\Python\que4.py"
Total number of words: 14
Unique words: {'handling', 'Dugani', 'we', 'see', 'in', 'Atharv', 'learn', 'that', 'file', 'python', 'will', 'Today'}
PS D:\career\Python>
```